Google Cloud SQL Database

Google Cloud SQL is a fully managed relational database service provided by **Google Cloud Platform (GCP)**. It allows users to deploy, manage, and scale popular relational databases such as **MySQL, PostgreSQL**, and **SQL Server** with minimal setup and maintenance. While "Google Cloud SQL Workspace" isn't a specific product, the term may refer to using Cloud SQL within a broader GCP development or operations environment—such as Cloud Workstations, Cloud Shell, or application development environments.

Use Cases for Google Cloud SQL

1. **Web and Mobile Applications**
   Cloud SQL is commonly used to power backends for web and mobile applications, including content management systems (e.g., WordPress), e-commerce platforms, and customer relationship management (CRM) systems.
2. **Business Applications**
   Host traditional business applications such as ERP, HR, or finance systems that require relational database backends.
3. **Data Warehousing and Analytics**
   Cloud SQL can serve as a staging or operational database before transferring data to BigQuery or other data warehouses for advanced analytics.
4. **SaaS Applications**
   Supports multi-tenant architectures for Software-as-a-Service solutions, providing high availability and scalability.
5. **Microservices Architectures**
   Acts as a persistent data store for containerized or Kubernetes-based microservices.
6. **Disaster Recovery and Backup**
   Provides automated backups, cross-region replication, and point-in-time recovery to meet disaster recovery and compliance needs.
7. **Development and Testing Environments**
   Enables developers to quickly provision databases for development, staging, or continuous integration environments.

Benefits of Google Cloud SQL

Benefit	Description
Fully Managed	Automates infrastructure, patching, replication, and backups.
High Availability	Regional configurations provide automatic failover with minimal downtime.
Scalability	Easily scale instances and storage based on workload demands.

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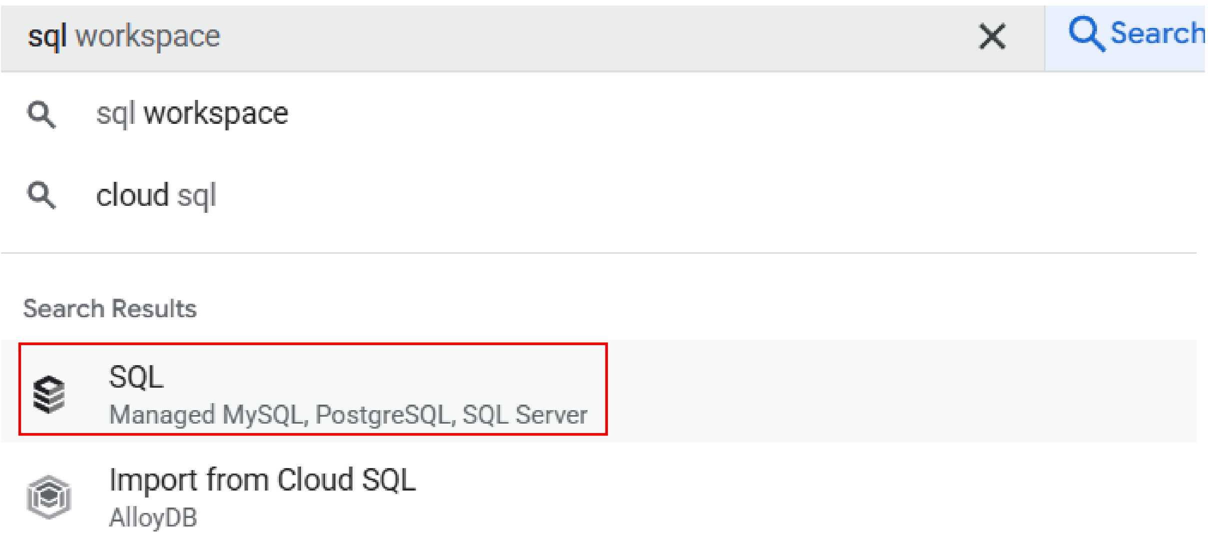
Security	Offers encryption at rest and in transit, IAM integration, private IP access, and VPC peering.
Automated Backups	Scheduled and on-demand backups with support for point-in-time recovery.
Performance Monitoring	Integrates with Cloud Monitoring and Query Insights for performance tuning and diagnostics.
Seamless Integration	Works natively with other GCP services such as BigQuery, Compute Engine, Dataflow, and App Engine.
Global Deployment	Deploy instances in multiple GCP regions to meet availability and latency requirements.

Complementary Tools for Workspace Integration

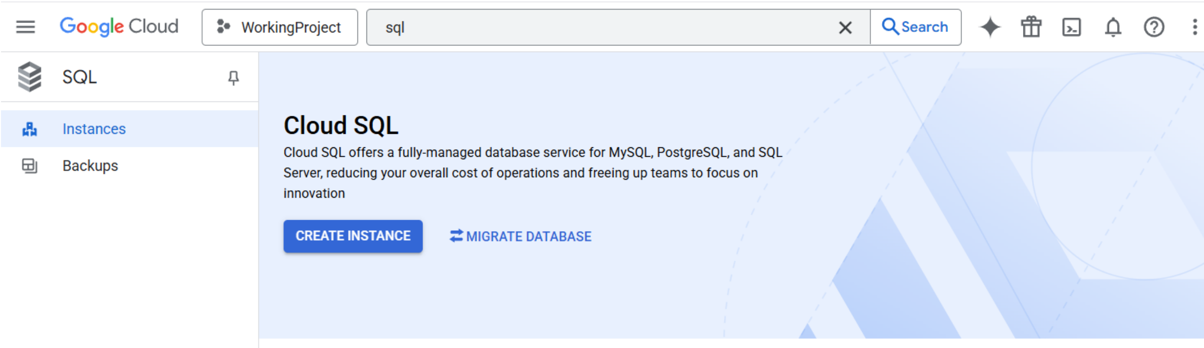
- **Cloud Shell or Cloud Workstations**: Secure, cloud-hosted development environments that connect easily to Cloud SQL.
- **Looker Studio or Dataform**: Analyze and transform data using SQL-based modeling tools.
- **Database Migration Service (DMS)**: Migrate databases from on-premise or other cloud providers with minimal downtime.
- **Terraform or Google Cloud Deployment Manager**: Define and manage Cloud SQL infrastructure as code.

To begin with the Lab

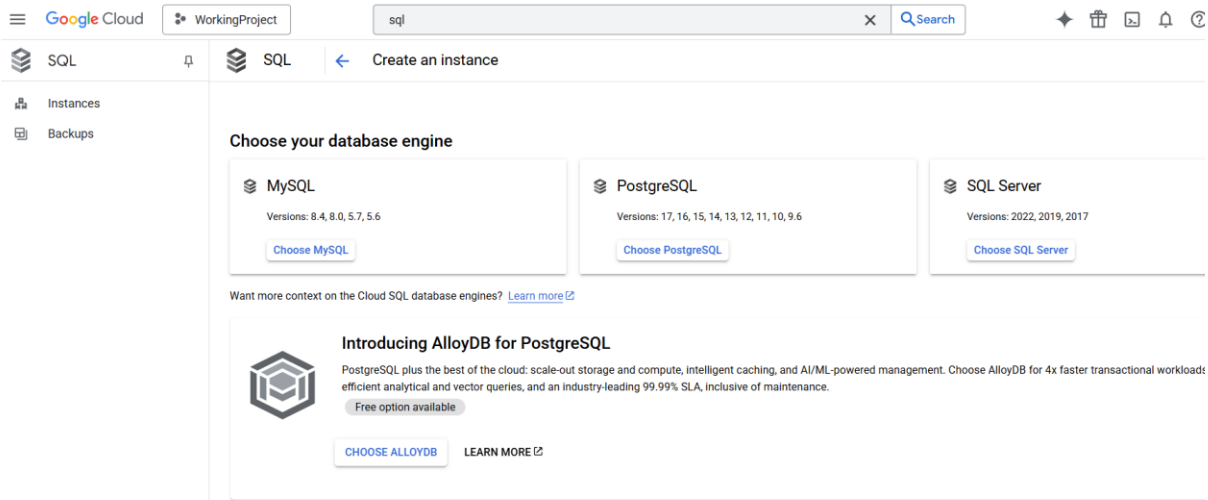
1. On your Google Cloud Console, search for SQL and choose the service accordingly.



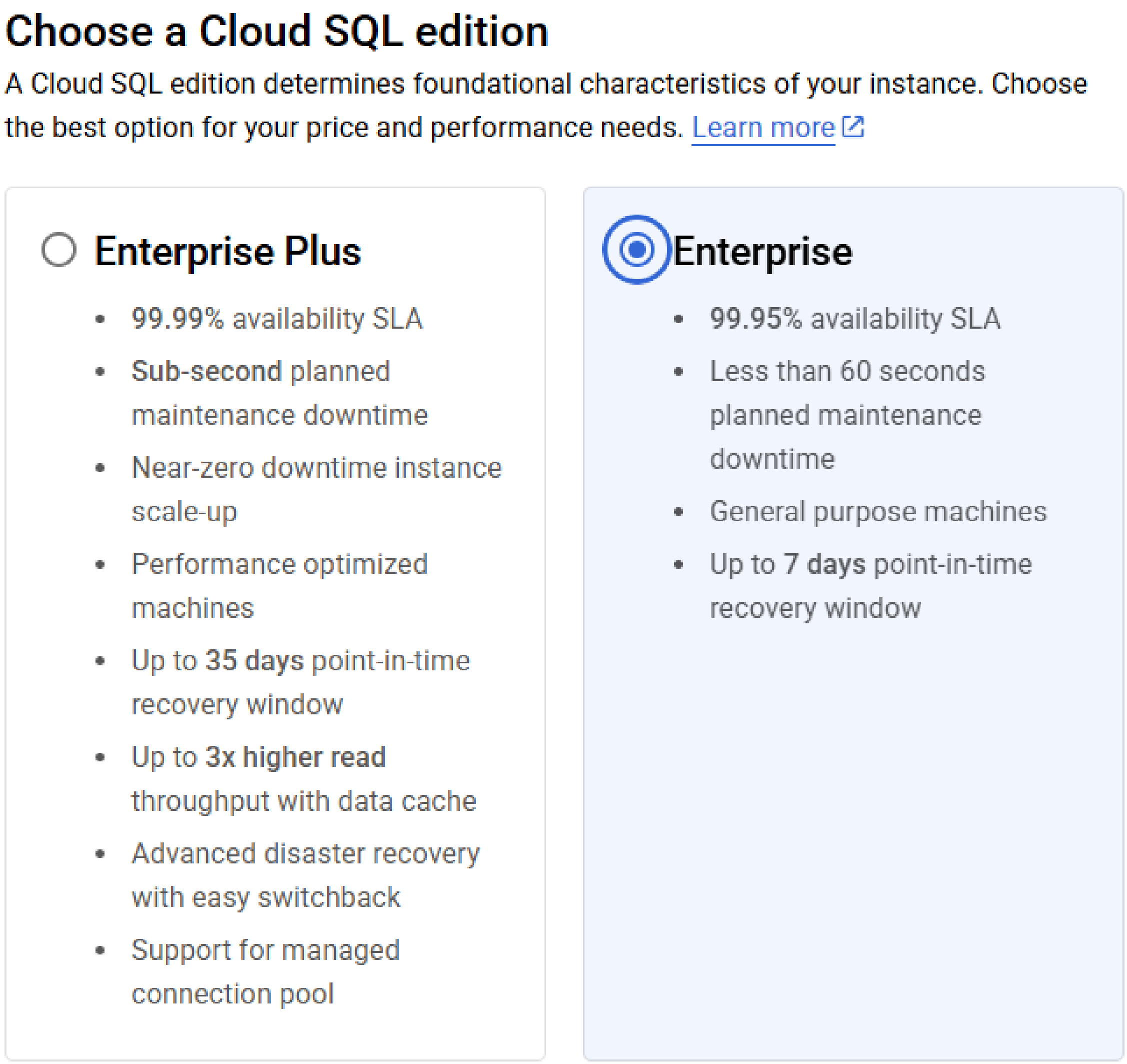
2. Now from the dashboard of Cloud SQL click on Create Instance.



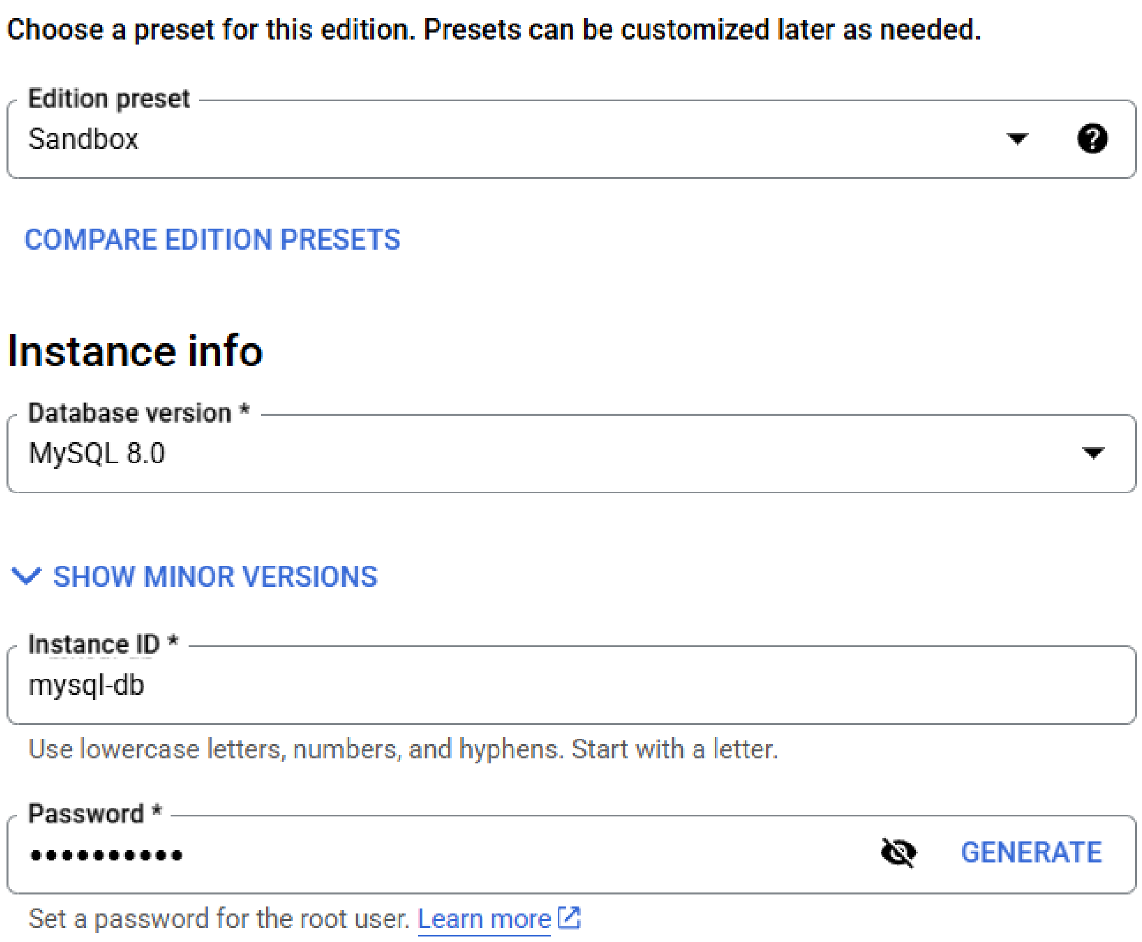
3. Then, you will see three database engines to choose from. You need to select MySQL. Click on choose and move forward.



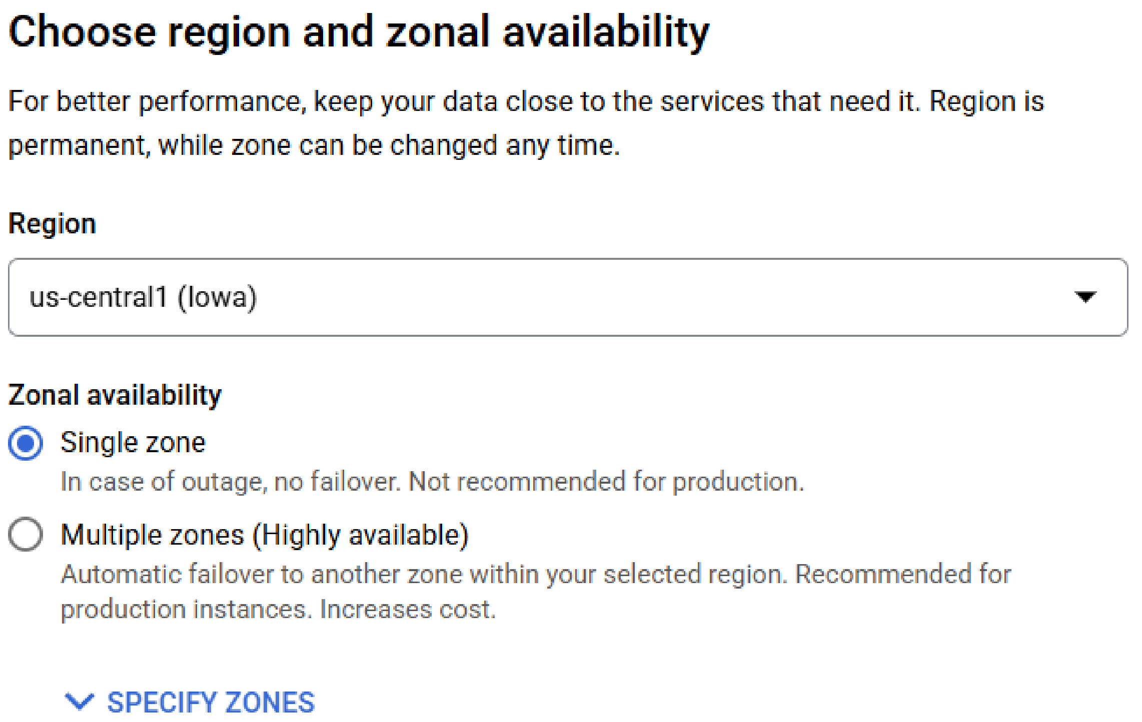
4. Now, for the first option, you must choose the Enterprise edition for this lab.



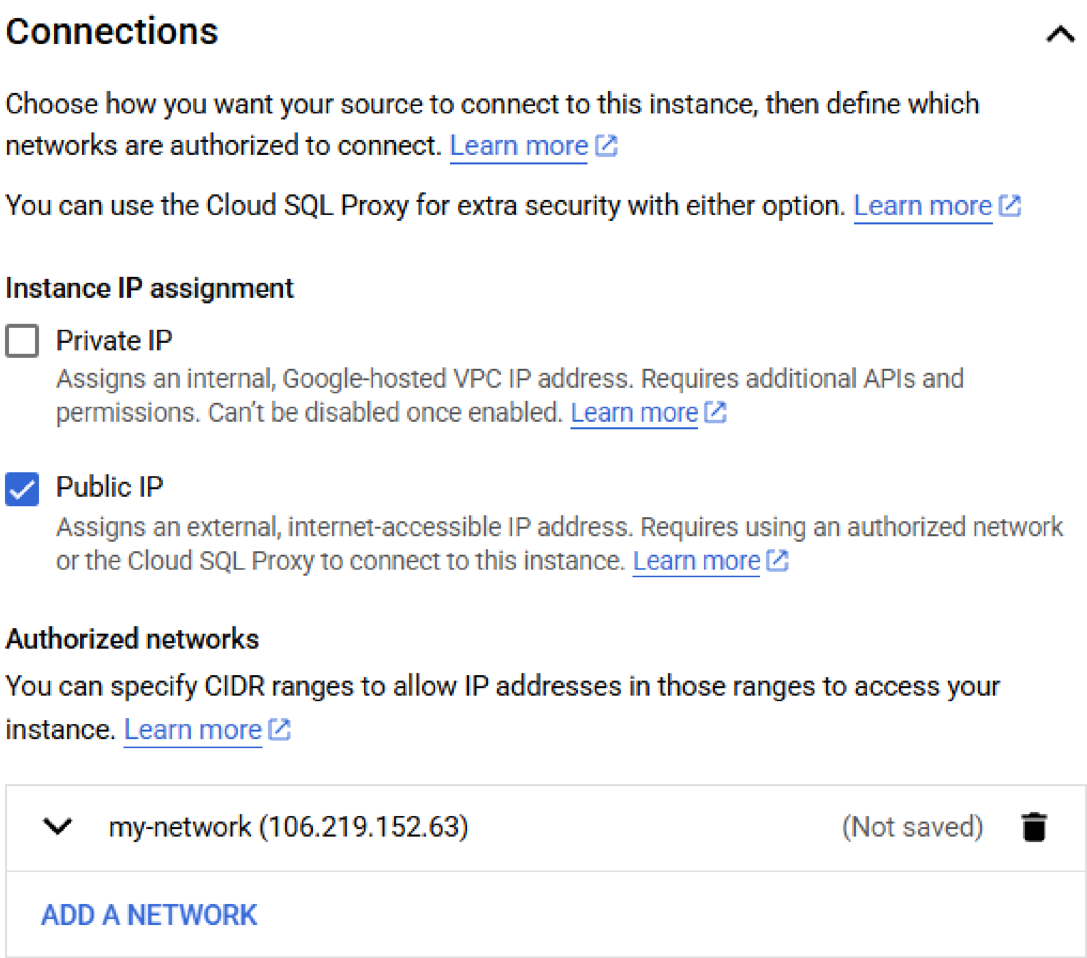
5. For the edition preset, choose Sandbox as it has the lowest number of CPUs and RAM, and it is much cheaper than the others, the best option for this lab.
6. Then you have to choose a database version; for now, leave it at the default.
7. In the end, just give a name and password to your instance.



8. After that, scroll down and choose a region and availability zone as per your preference.



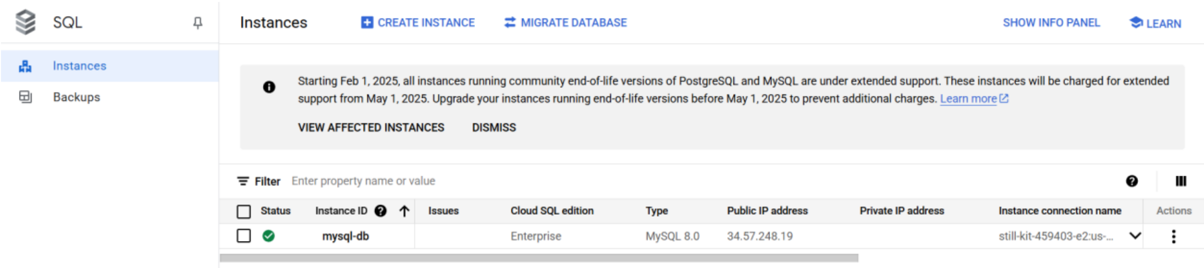
9. Now scroll down to connections and you will see that your DB instance has a pubic IP that will be attached to it while creation.
10. So, now we need to add an authorized network so that we can establish a connection with our DB instance.
11. For that, click on add a network, give a name to your network and then write the IP of your local machine.



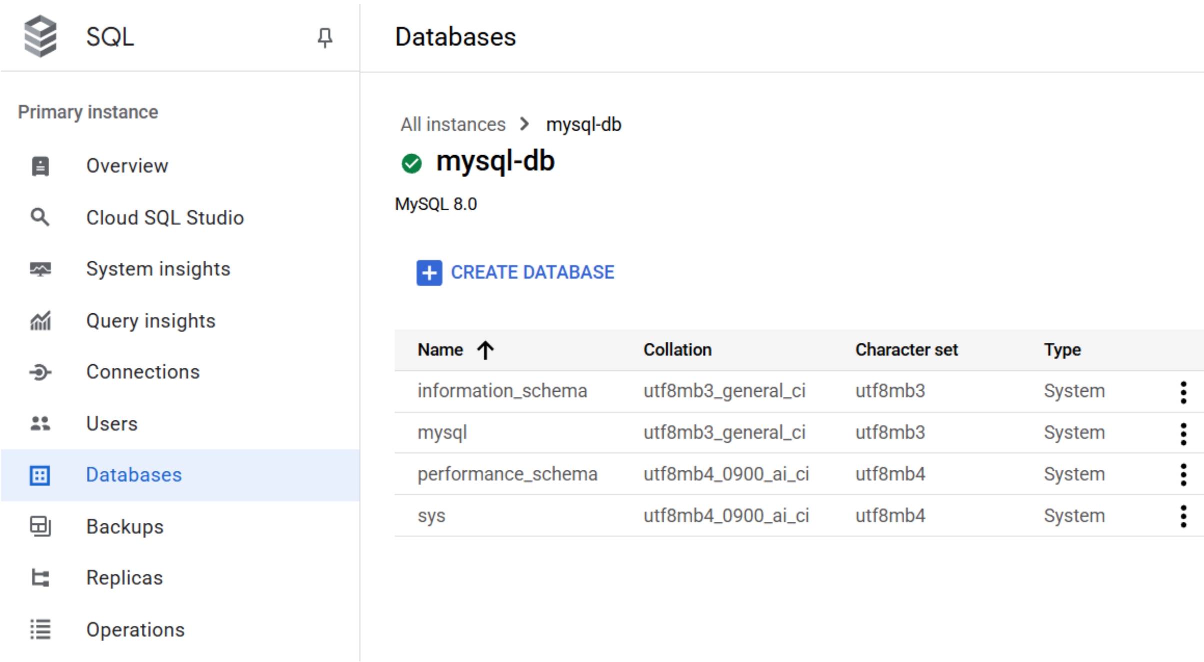
12. In the end, just click on create instance, or you can explore the other options for your knowledge, but do not change anything, and click on create.
13. Now wait for your instance to get created, or what you can do is visit the link given below to download MySQL Workbench.

    https://www.mysql.com/products/workbench/

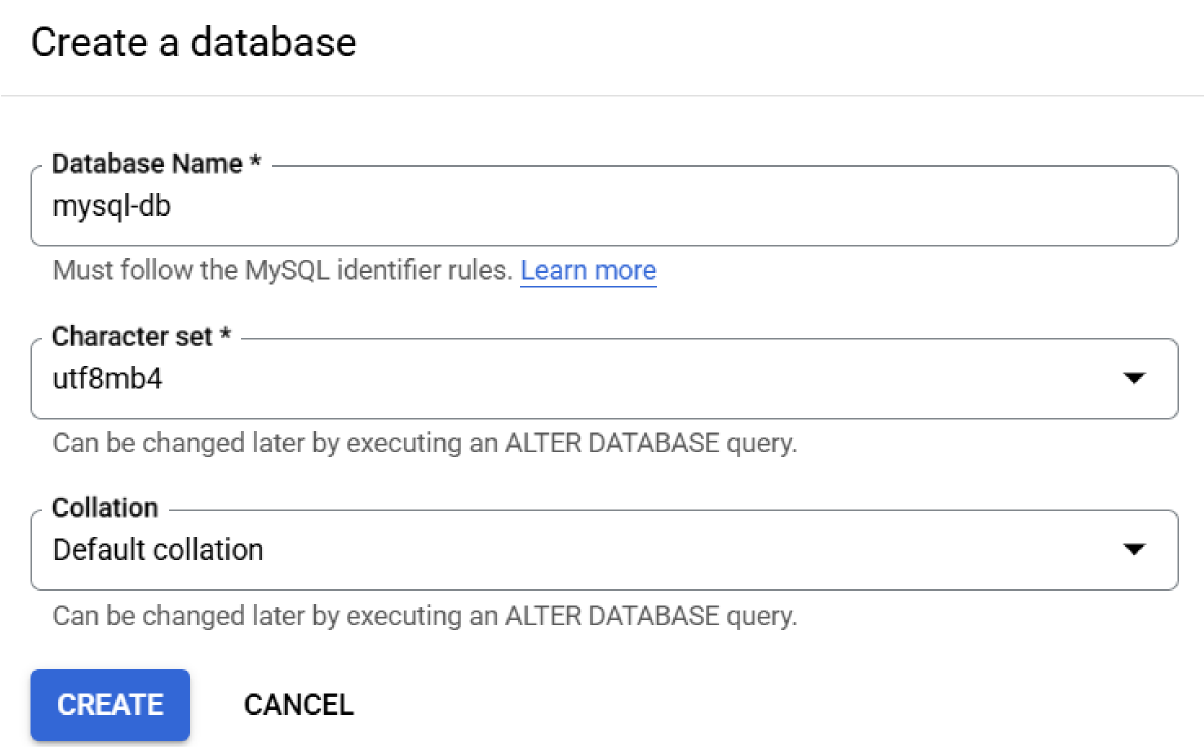
14. Here you can see that our instance has been created, now click on it to go inside.



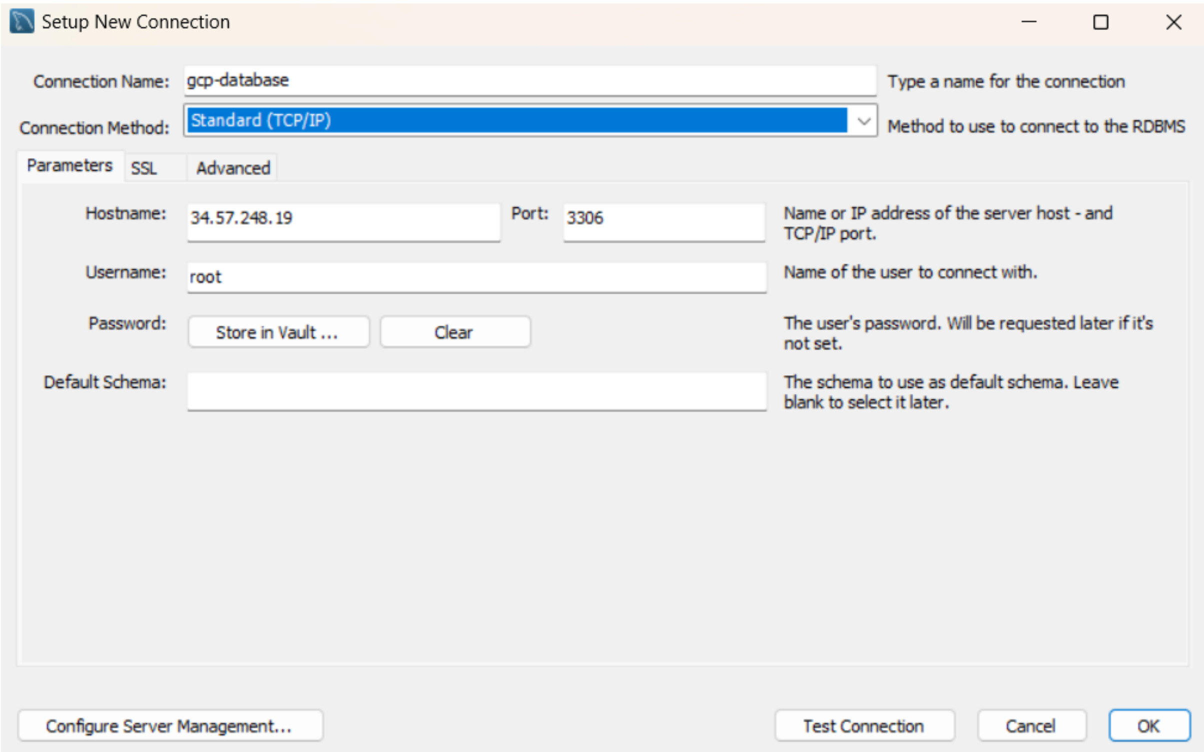
15. As you can see here, we have some default databases that have been created by default. But we will create our own database, so click on Create DB.



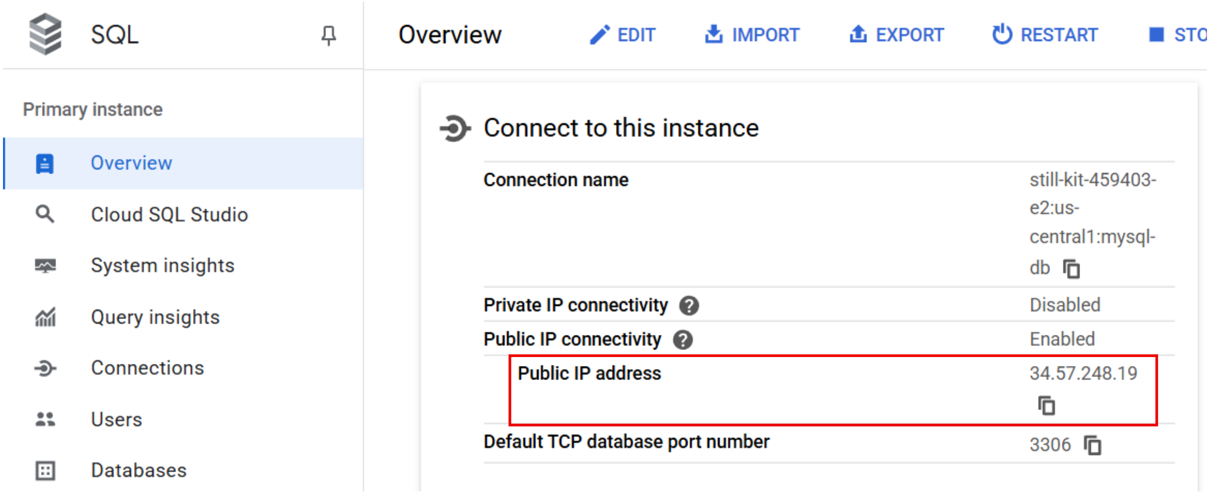
16. To create a Database, you just have to give a name and click on Create.



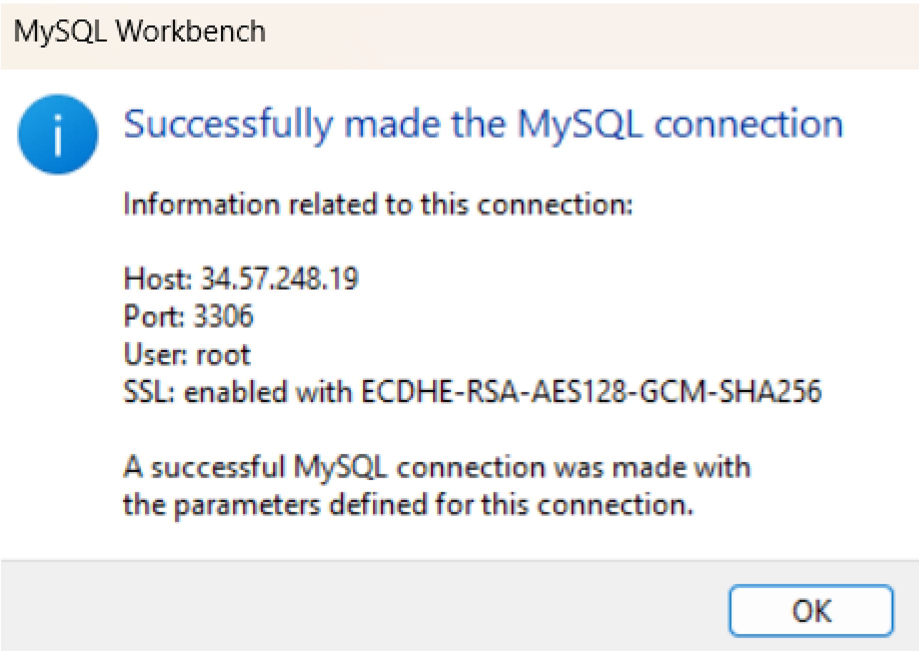
17. After creating your Database, open the MySQL Workbench tool and create a new connection there.
18. Give a connection name, and then for the host name you have to give the public IP of your database instance.



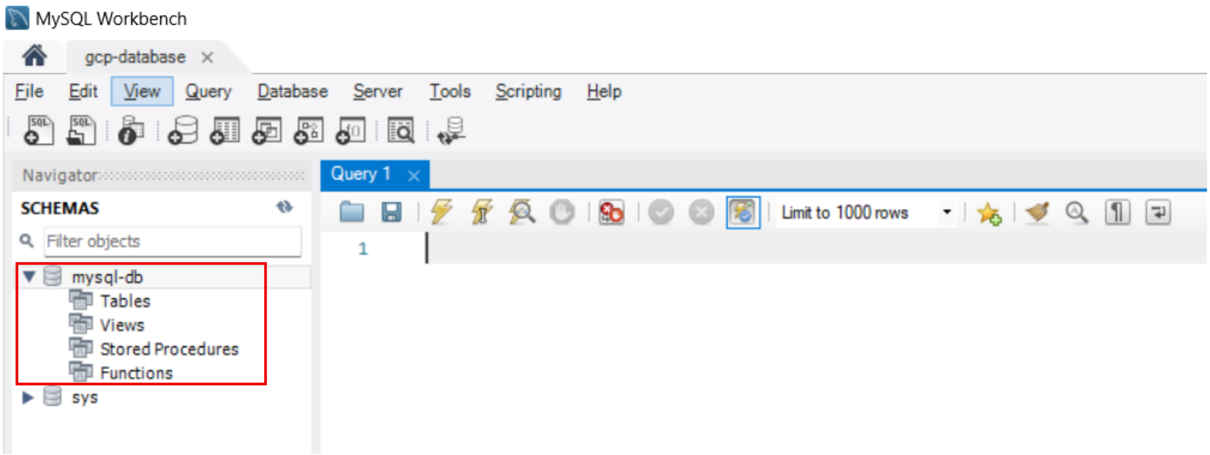
19. To get the public IP address of your DB instance, you need to go back to your GCP console.
20. From the overview page of your instance, you can get the Public IP address. Copy it.



21. Now you need to click on test connection first before actually connecting with your DB instance.
22. So, click on Test connection. Then you need to give the password that you defined while creating it. And click on test. You will see a window that says Connection successful.
23. Then click on OK and go inside your Database.



24. And here you can see your Database.



25. Now you can use it to run the basic SQL commands and have a basic understanding of your DB.
26. In the below snapshot, we used our database and then created a table, inserted some data init and then we displayed the data.